Physical Chemistry (I)

Lecture 11

Entropy: Statistical View

In the Last Class

- Entropy and the second law
  - Clausius inequality and the second law
  - Fundamental equation
  - Entropy calculation
- The third law and thermochemistry
  - Third law
  - Thermochemistry of entropy
- Thermodynamic meaning of entropy

Last class

Entropy of Surroundings

The surroundings are huge, and their pressure, volume, and temperature are invariant:

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T_{\text{sur}}}$$

Clausius Inequality

$$\mathrm{d}S \geq \frac{\delta q}{T}$$

3rd version of the second law of thermodynamics:

"The entropy of an isolated system never decreases."

Fundamental Equation

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

As $U = U(S, V)$ is a state function,

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_V \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_S \mathrm{d}V$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S}\right)_V, \qquad p = -\left(\frac{\partial U}{\partial V}\right)_S$$

Examples

Heat transfer

Expansion (reversible vs. free)

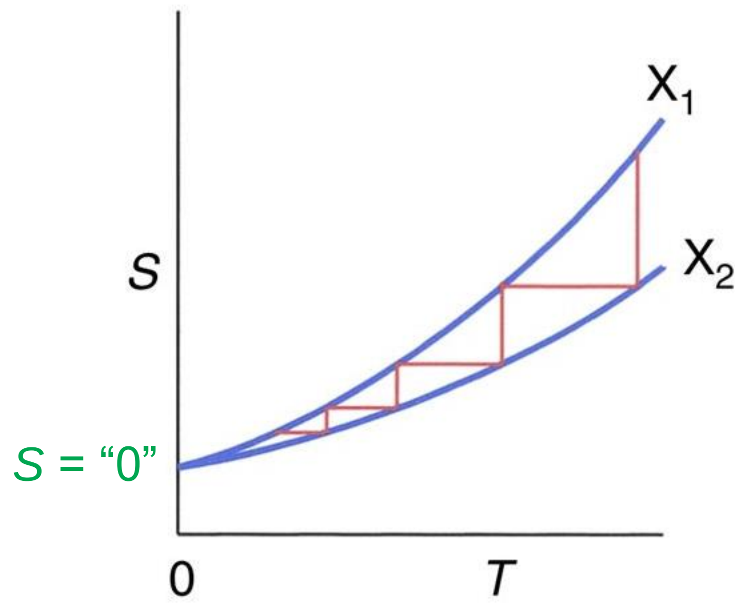
Phase transitions

Heating

Last class

Third Law of Thermodynamics

"The entropy of all perfect crystalline substances is zero at T = 0."

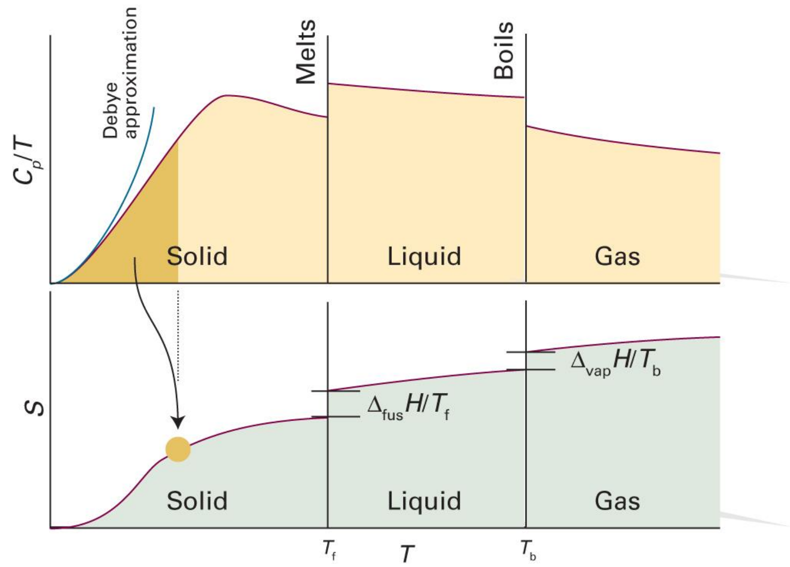


(Image: Masanes and Oppenheim, *Nature Communications* **8**: 14538 (2017), Figure 1)

Last class

Third-Law Entropy

$$S(T) = \int_0^T \frac{C_p(T')}{T'} \mathrm{d}T'$$



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 3C.1)

Last class

Thermochemistry of Entropy

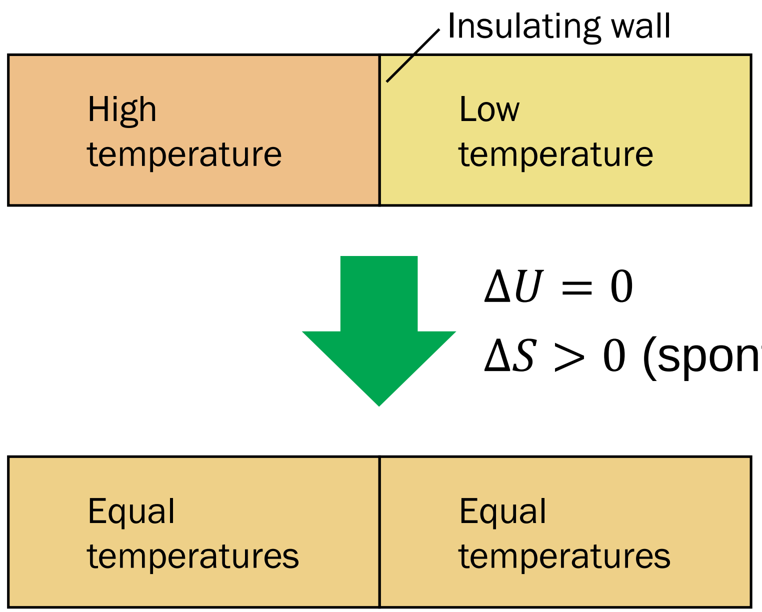
- Standard entropy $S^{\ominus}$: third-law entropy of a pure substance at 1 bar (always positive)

- Molar standard entropy $S_{\mathrm{m}}^{\ominus} = S^{\ominus}/n$

- Standard reaction entropy $\Delta_{\mathrm{r}}S^{\ominus}$

$$\Delta_{\mathrm{r}}S^{\ominus} = \sum_{\text{products } i} \nu(i)S_{m}^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j)S_{m}^{\ominus}(j)$$

Entropy and Work



$\Delta U = 0$

$\Delta S > 0$ (spontaneous)

"Index of Exhaustion"

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Statistical Consideration

- (Macro)state: defined by values of macroscopic variables
- Microstate: specific microscopic arrangement
- Different microstates can be mapped into the same macrostate

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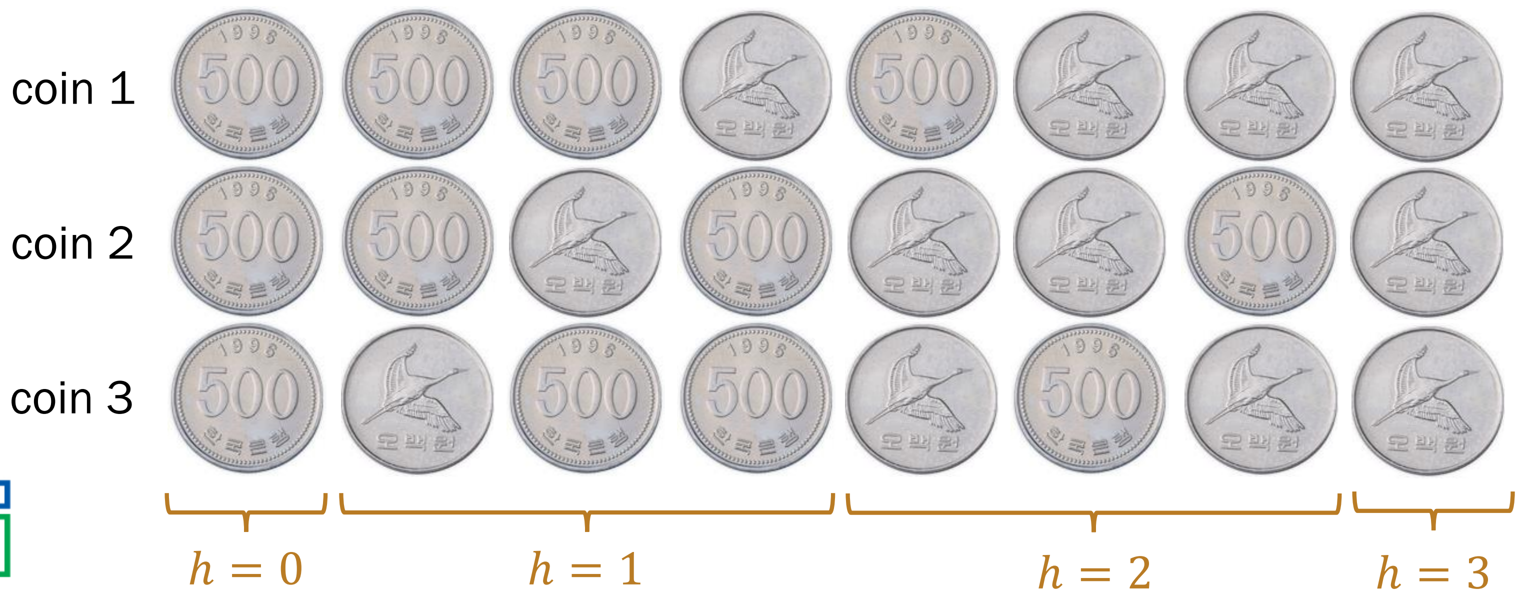
Example: Coin Toss

When tossing three coins,

- Macrostate: defined by the number of heads h
- Microstate: the actual arrangement of coins



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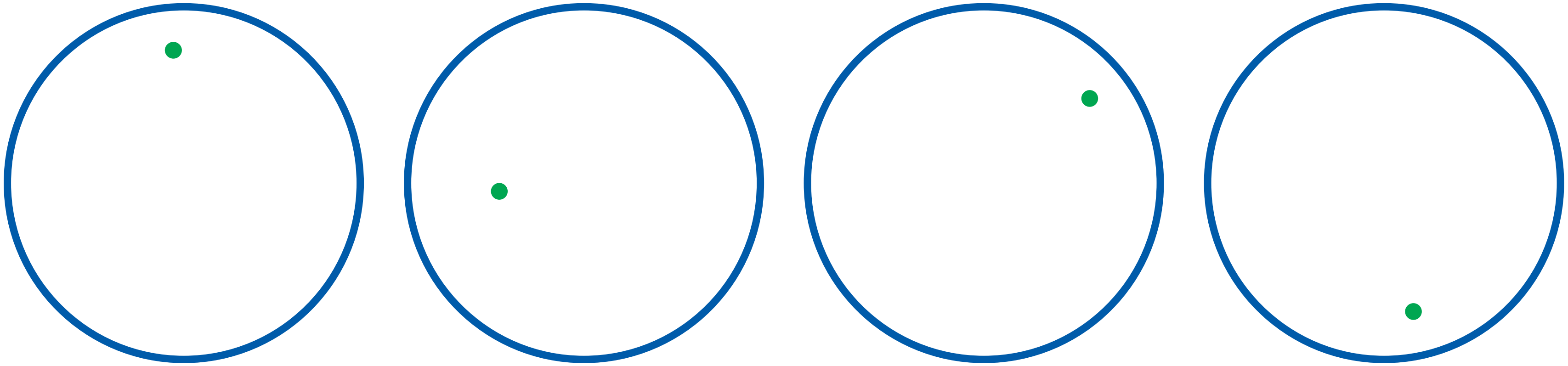
Example: Diffusion

When a gas molecule moves around the system,

- Macrostate: defined by the volume V
- Microstate: the position of the molecule



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Number of Microstates *W*

"The number of microstates

corresponding to a specific macrostate"

- Coin toss

$W(h = 0) = 1, W(h = 1) = 3, W(h = 2) = 3, W(h = 3) = 1$

- Diffusion

Let the volume occupied by a molecule v_0;

for the system volume V, $W(V) = V/v_0$

Boltzmann's Formula

For an isolated system with fixed internal energy,

$$S = k_{\mathrm{B}} \ln W$$

number of microstates

Boltzmann constant (1.38×10^{-23} J/K)

$$\Delta S = S_f - S_i = k_{\mathrm{B}} \ln W_f - k_{\mathrm{B}} \ln W_i = k_{\mathrm{B}} \ln\left(\frac{W_f}{W_i}\right)$$

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Example: Diffusion

Suppose that the system volume changes from V_i to V_f:

$$W_i = V_i / v_0$$

$$W_f = V_f / v_0$$

The change in entropy is given by

$$\Delta S_1 = k_\mathrm{B} \ln\left(W_f / W_i\right) = k_\mathrm{B} \ln\left(V_f / V_i\right)$$

For a perfect gas consisting of N molecules,

$$\Delta S_N = N k_\mathrm{B} \ln\left(V_f / V_i\right) = nR \ln\left(V_f / V_i\right)$$

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Second Law Revisited

$$\mathrm{d}S \geq 0 \text{ (for an isolated system)}$$

As $S = k_\mathrm{B} \ln W$ is a monotonic function of W,

$$\mathrm{d}W \geq 0 \text{ (for an isolated system)}$$

The number of microstates never decreases

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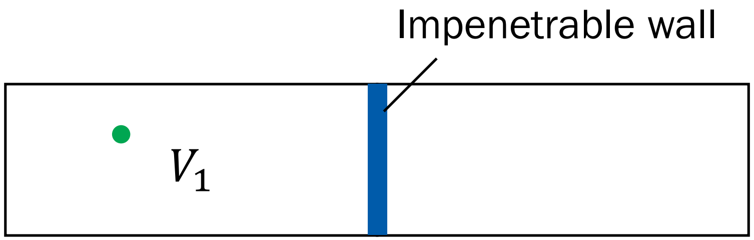
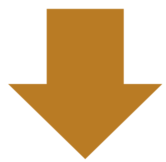
Example: Diffusion



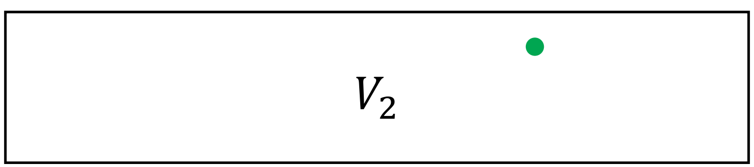
$W_1 = V_1 / v_0$

$S_1 = k_\mathrm{B} \ln W_1$



$\Delta W > 0$

$\Delta S > 0$ (spontaneous)



$W_2 = V_2 / v_0$

$S_2 = k_\mathrm{B} \ln W_2$

Third Law Revisited

At $T = 0,$

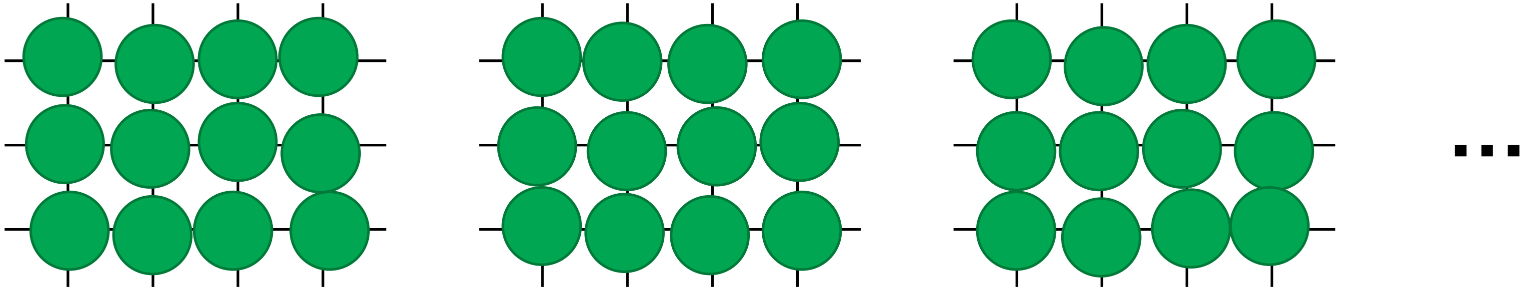
$$S = k_{\mathrm{B}} \ln W = 0$$

$$W = 1$$

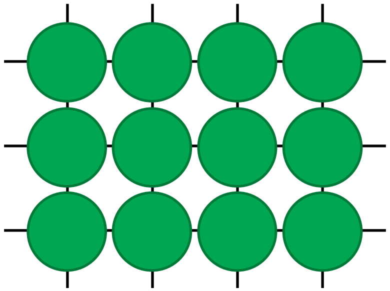
Only one microstate is possible at 0 K

Solid at Absolute Zero

$$T > 0$$



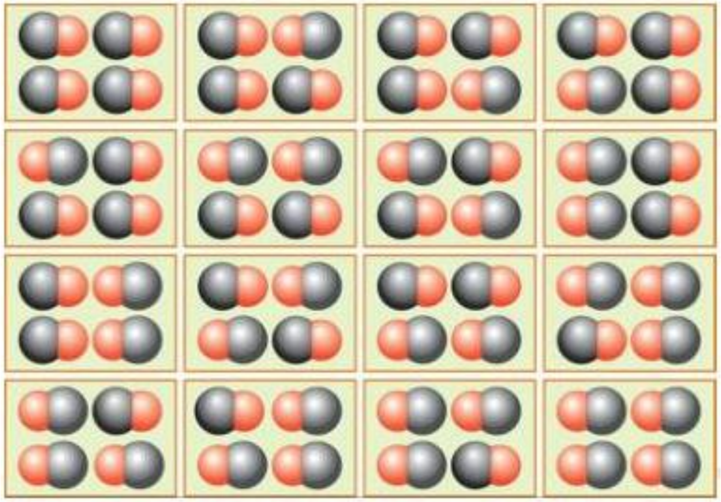
$$T = 0$$



Residual Entropy

$W > 1$ and $S > 0$ even at $T = 0$

e.g. carbon monoxide (CO)



(Image: https://image2.slideserve.com/4094967/slide18-l.jpg)

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Spins

Electrons and protons have spins: up (↑) or down (↓)

Consider a solid system consisting of N spins:

- Two spins: 4 microstates (↑↑, ↑↓, ↓↑, ↓↓)
- Three spins: 8 microstates
- N spins: 2^N microstates

Similar!

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Spins in a Magnetic Field

The two spins react to a magnetic field differently:

$$E(\uparrow) = +\frac{\epsilon}{2}, \qquad E(\downarrow) = -\frac{\epsilon}{2}$$

The internal energy of the system is

$$U = \frac{\epsilon}{2}N_{\uparrow} - \frac{\epsilon}{2}N_{\downarrow} = \frac{\epsilon}{2}(N_{\uparrow} - N_{\downarrow})$$

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Macrostate

- Total number of spins $N = N_\uparrow + N_\downarrow$

- Internal energy

$$U = \frac{\epsilon}{2}(N_\uparrow - N_\downarrow) = \epsilon N_\uparrow - \frac{\epsilon}{2}N$$

- Macrostate: number of “up” spins

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Entropy

For a given macrostate (number of up spins = M), what is the number of microstates?

M spins are "up" among N spins: combinations

$$W = {}_N C_M = \frac{N!}{M!\,(N-M)!}$$

$$S = k_\mathrm{B} \ln W = k_\mathrm{B} \ln \left[\frac{N!}{M!\,(N-M)!} \right]$$

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Stirling's Approximation

$\ln N! \approx N \ln N - N$ (for large N)

$$\begin{aligned}
\ln N! &= \ln\{N \times (N-1) \times (N-2) \times \cdots\} \\
&= \ln N + \ln(N-1) + \ln(N-2) + \cdots \\
&= \textstyle\sum_{k=1}^{N} \ln k \\
&\approx \textstyle\int_1^N \ln x \, dx \\
&= [x \ln x - x]_1^N \\
&= N \ln N - N + 1 \\
&\approx N \ln N - N \text{ (for large } N \gg 1\text{)}
\end{aligned}$$

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Simplification

$$S = k_\mathrm{B} \ln \left[\frac{N!}{M!\,(N-M)!} \right]$$

$$= k_\mathrm{B}[\ln N! - \ln M! - \ln(N-M)!]$$

Using Stirling's approximation,

$$S = k_\mathrm{B}[N \ln N - M \ln M - (N-M)\ln(N-M)]$$

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Entropy and Internal Energy

$$S = k_{\mathrm{B}}[N \ln N - M \ln M - (N - M) \ln(N - M)]$$

For a system with M up spins,

$$U = \epsilon M - \frac{\epsilon}{2} N$$

$$M = \frac{N}{2} + \frac{U}{\epsilon}$$

$$S = k_{\mathrm{B}}\Big[N \ln N - \left(\frac{N}{2} + \frac{U}{\epsilon}\right) \ln\left(\frac{N}{2} + \frac{U}{\epsilon}\right) - \left(\frac{N}{2} - \frac{U}{\epsilon}\right) \ln\left(\frac{N}{2} - \frac{U}{\epsilon}\right)\Big]$$

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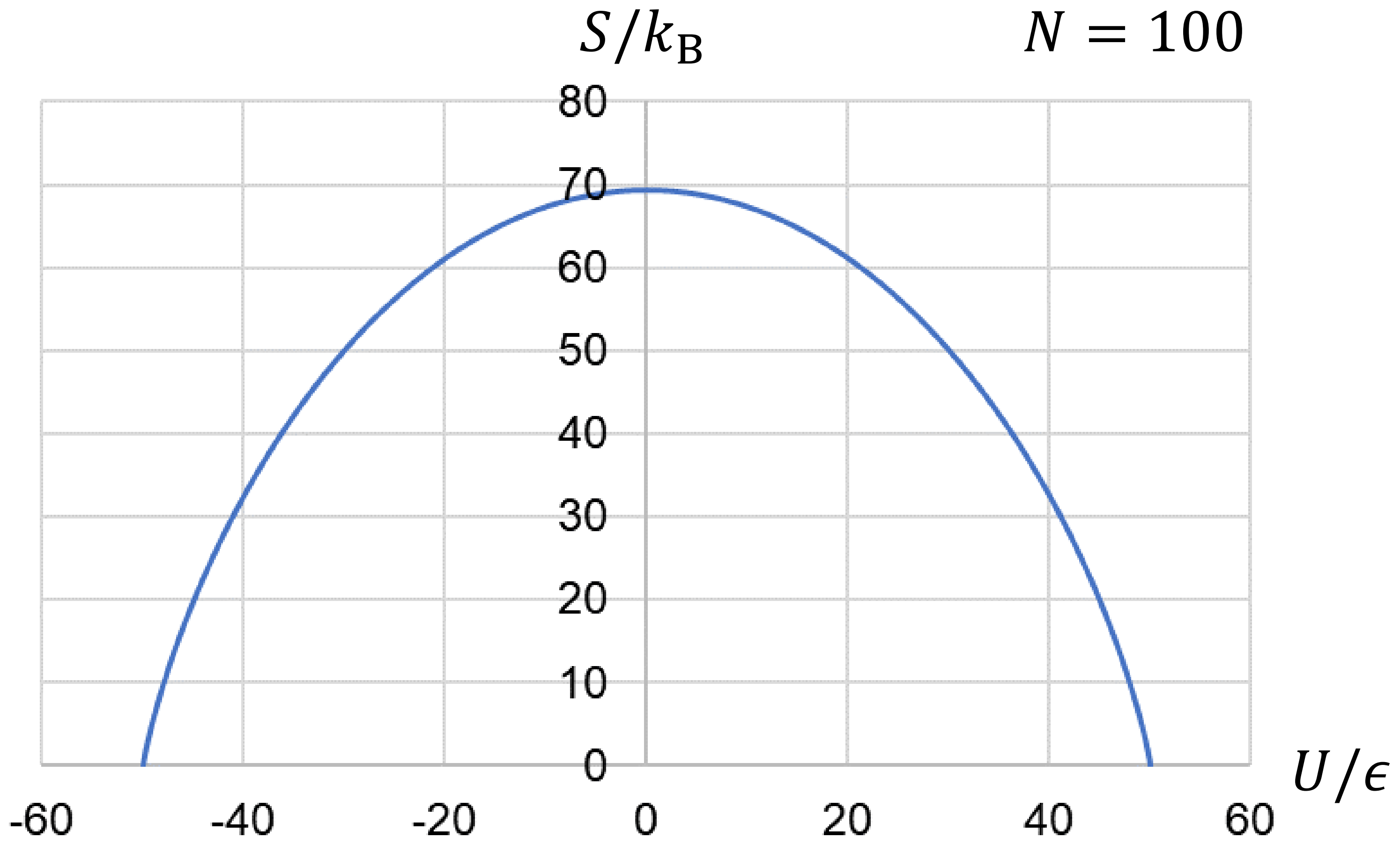
$\boldsymbol{S(U)}$



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Temperature

From the fundamental equation of thermodynamics,

$$T = \left(\frac{\partial U}{\partial S}\right)_V, \qquad \frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_V$$

$$S = k_{\mathrm{B}}\left[N \ln N - \left(\frac{N}{2} + \frac{U}{\epsilon}\right)\ln\left(\frac{N}{2} + \frac{U}{\epsilon}\right) - \left(\frac{N}{2} - \frac{U}{\epsilon}\right)\ln\left(\frac{N}{2} - \frac{U}{\epsilon}\right)\right]$$

$$\frac{1}{T} = \frac{k_{\mathrm{B}}}{\epsilon}\ln\left(\frac{N}{2} - \frac{U}{\epsilon}\right) - \frac{k_{\mathrm{B}}}{\epsilon}\ln\left(\frac{N}{2} + \frac{U}{\epsilon}\right) = \frac{k_{\mathrm{B}}}{\epsilon}\ln\left(\frac{N/2 - U/\epsilon}{N/2 + U/\epsilon}\right)$$

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Meaning of Temperature

$$\frac{1}{T} = \frac{k_\mathrm{B}}{\epsilon} \ln\left(\frac{N/2 - U/\epsilon}{N/2 + U/\epsilon}\right)$$

- Number of up spins: $N_\uparrow = M = N/2 + U/\epsilon$
- Number of down spins: $N_\downarrow = N - M = N/2 - U/\epsilon$

$$\frac{1}{T} = \frac{k_\mathrm{B}}{\epsilon} \ln\left(\frac{N_\downarrow}{N_\uparrow}\right)$$

- If $N_\downarrow > N_\uparrow$, the temperature is positive
- If $N_\downarrow < N_\uparrow$, the temperature is negative

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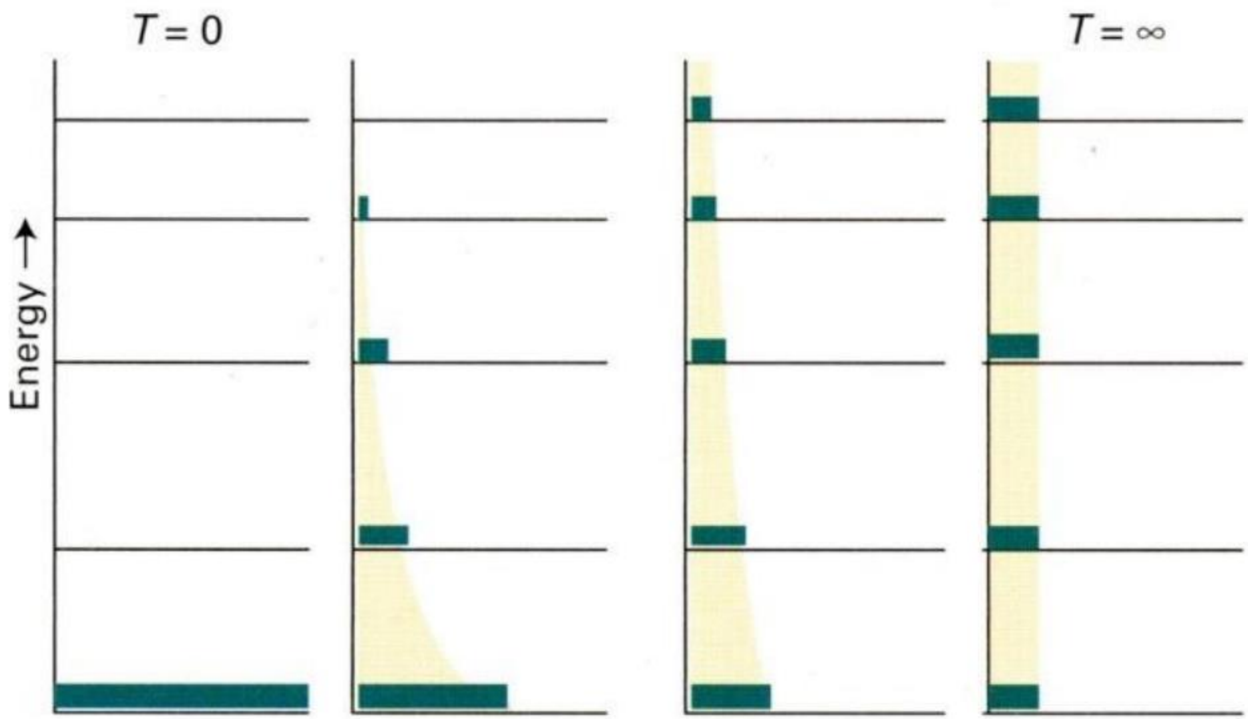
Temperature and Populations

$P(E) \propto e^{-E/k_\mathrm{B}T}$ (Boltzmann factor)



(Image: https://www.chem.tamu.edu/class/majors/chem328/CHEM%20328_Chapter%2015.pdf)

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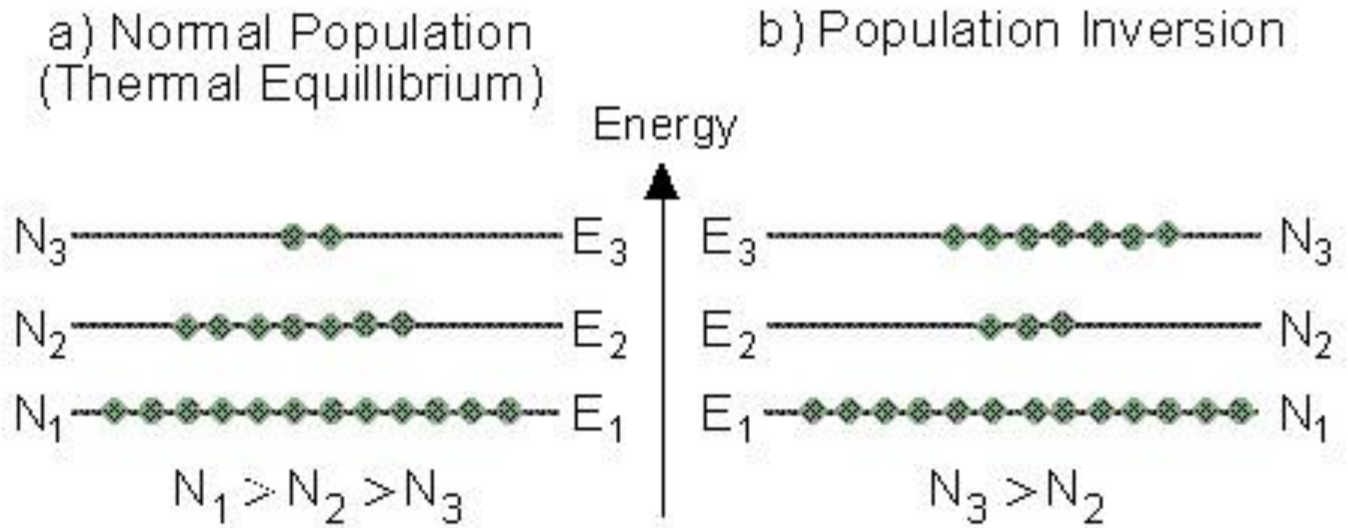
Population Inversion



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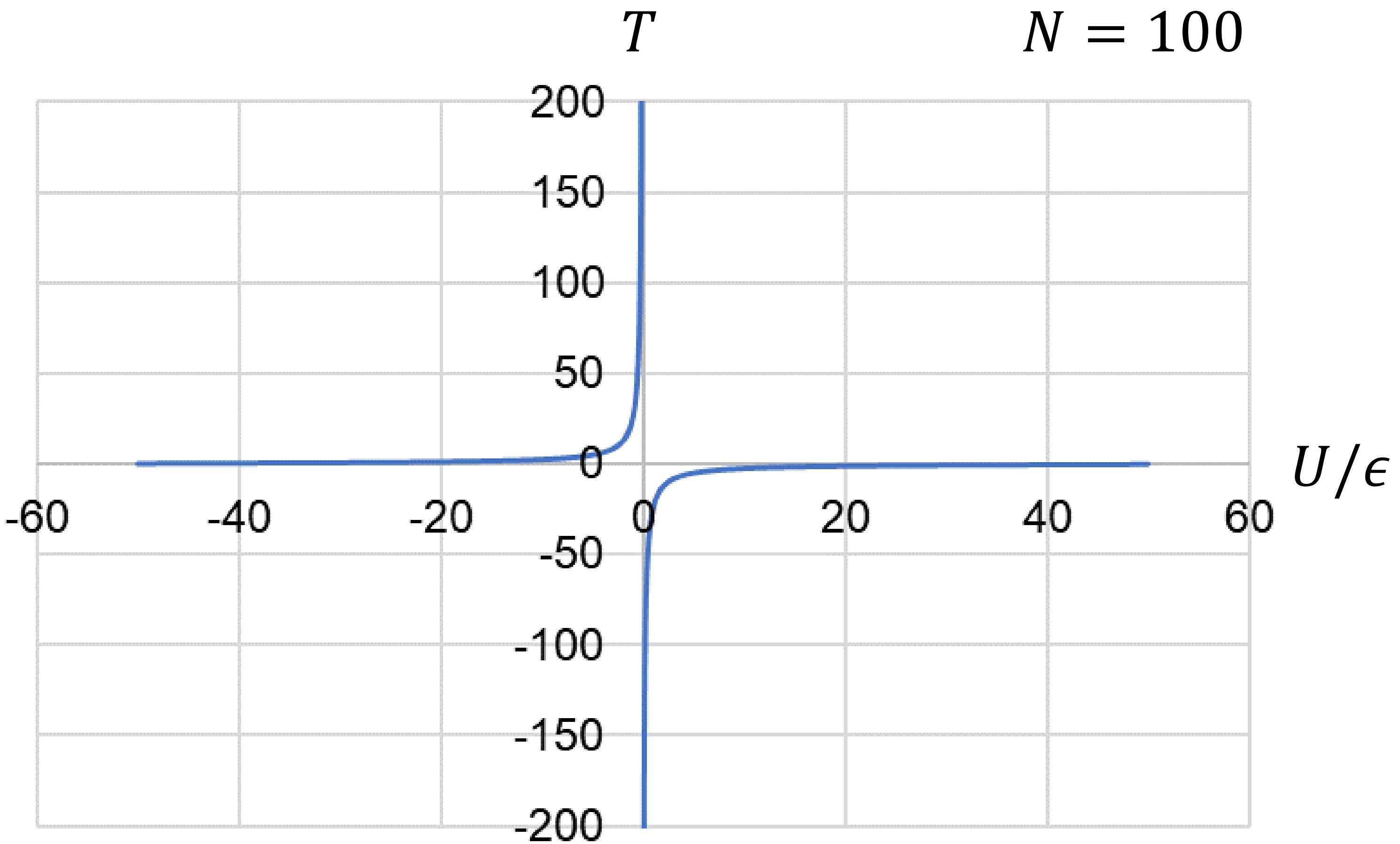
$T(U)$



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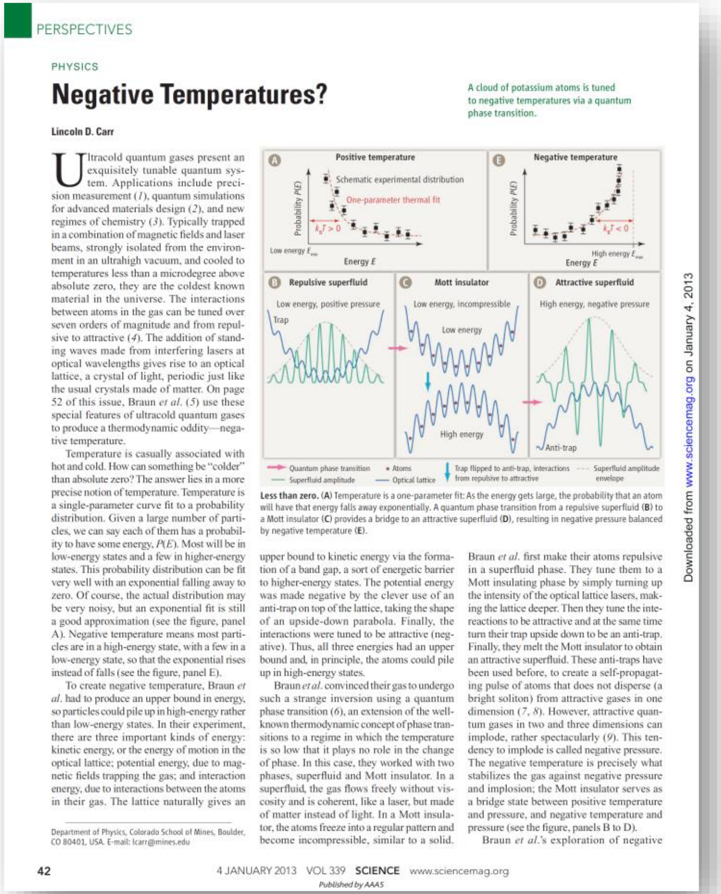
Measuring Negative T


PERSPECTIVES

PHYSICS

Negative Temperatures?

Lincoln D. Carr

A cloud of potassium atoms is tuned to negative temperatures via a quantum phase transition.

Less than zero. (A) Temperature is a one-parameter fit: As the energy gets large, the probability that an atom will have that energy falls away exponentially. A quantum phase transition from a repulsive superfluid (B) to a Mott insulator (C) provides a bridge to an attractive superfluid (D), resulting in negative pressure balanced by negative temperature (E).

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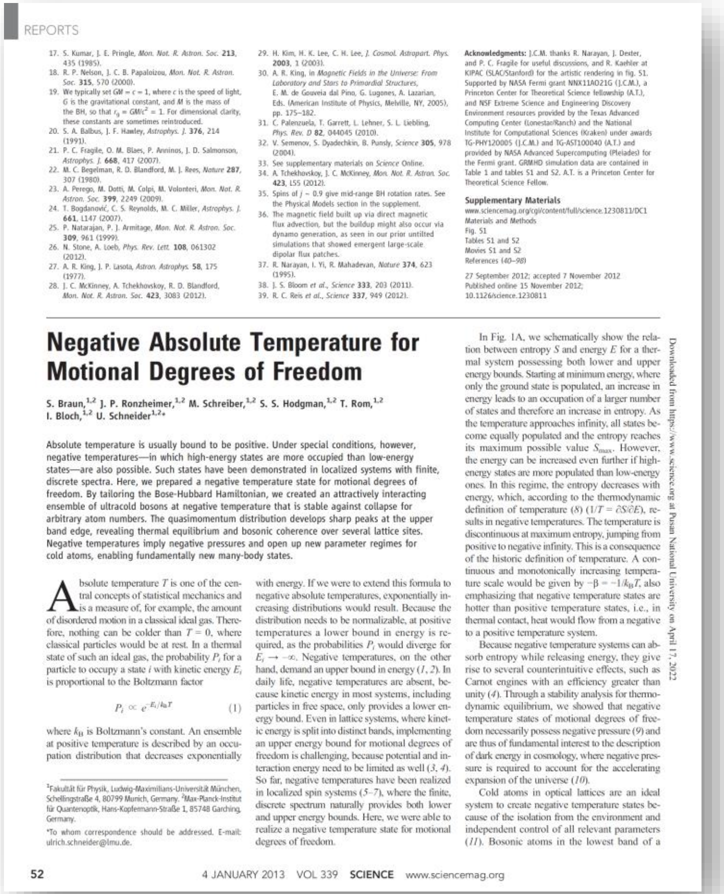
REPORTS

Negative Absolute Temperature for Motional Degrees of Freedom

S. Braun,[1,2] J. P. Ronzheimer,[1,2] M. Schreiber,[1,2] S. S. Hodgman,[1,2] T. Rom,[1,2] I. Bloch,[1,2] U. Schneider[1,2]*

Absolute temperature is usually bound to be positive. Under special conditions, however, negative temperatures—in which high-energy states are more occupied than low-energy states—are also possible. Such states have been demonstrated in localized systems with finite, discrete spectra. Here, we prepared a negative temperature state for motional degrees of freedom. By tailoring the Bose-Hubbard Hamiltonian, we created an attractively interacting ensemble of ultracold bosons at negative temperature that is stable against collapse for arbitrary atom numbers. The quasimomentum distribution develops sharp peaks at the upper band edge, revealing thermal equilibrium and bosonic coherence over several lattice sites. Negative temperatures imply negative pressures and open up new parameter regimes for cold atoms, enabling fundamentally new many-body states.

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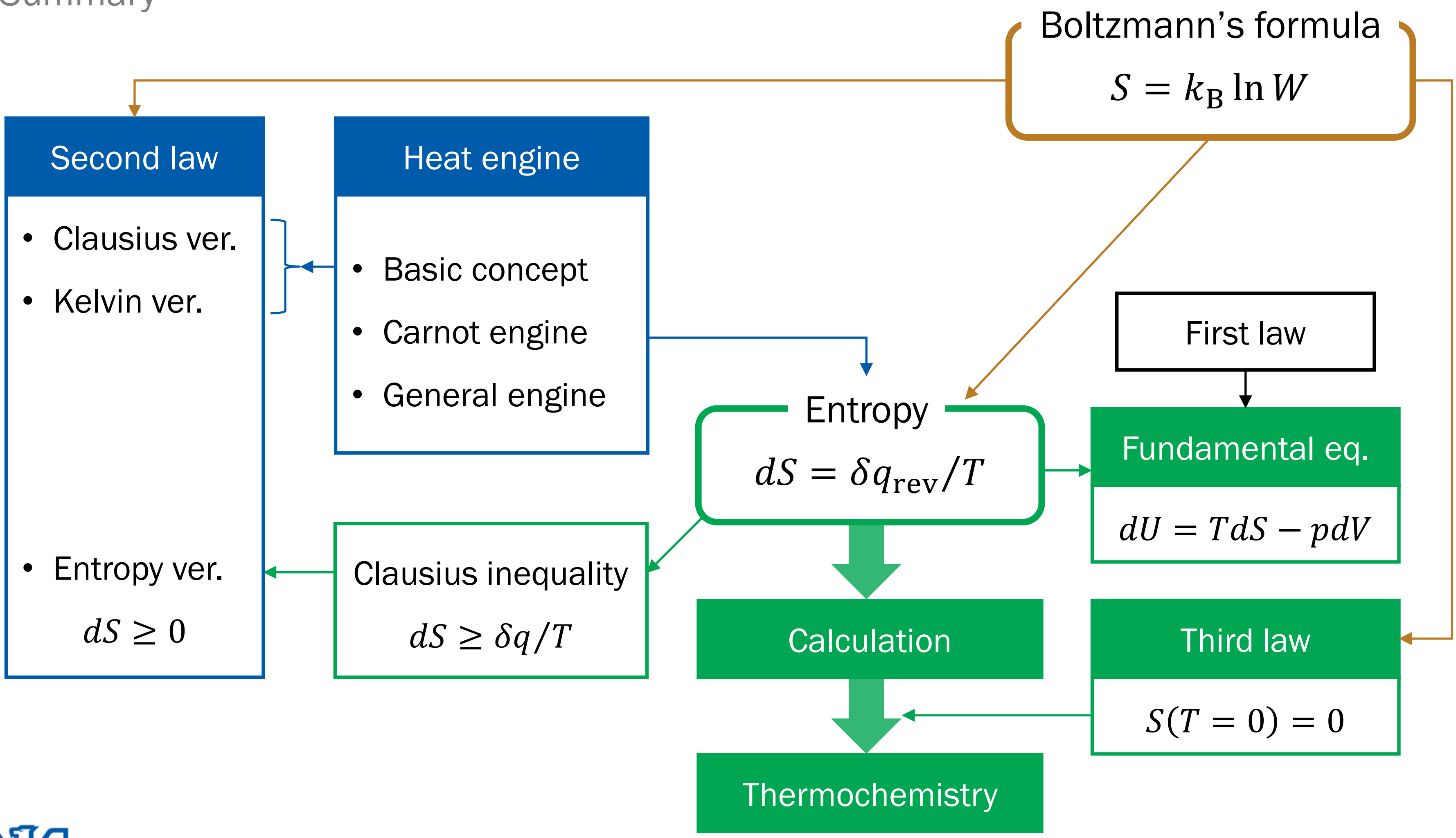
Science (2013)

Summary



Boltzmann's formula
$S = k_{\mathrm{B}} \ln W$
Second law
Clausius ver.
Kelvin ver.
Entropy ver.
$dS \geq 0$
Heat engine
Basic concept
Carnot engine
General engine
Clausius inequality
$dS \geq \delta q/T$
Entropy
$dS = \delta q_{\mathrm{rev}}/T$
First law
Fundamental eq.
$dU = TdS - pdV$
Calculation
Third law
$S(T = 0) = 0$
Thermochemistry